Sahaj Bhadja

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Summary

Software Engineer with an M.S. in Computer Science (3.75 GPA) and B.Tech (3.88 GPA), and 2+ years building **C++**, Python, and Java systems on **Linux**, with concurrency, networking, automated testing, and production debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **concurrent job processing** with worker processes, explicit job-state machines, and failure isolation, processing 1K+ jobs at 3x throughput without cascading failures under heavy concurrent load.
- **Profiled and optimized** data access for a 500+ user platform, cutting latency from 1.2s to 350ms, and monitored integration jobs to sustain 99%+ reliability through CI/CD releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a request **routing** layer across multiple inference backends with dynamic switching, timeouts, and fallbacks, keeping service available for 10K+ daily requests from 150+ instructors on AWS.
- Designed a **multithreaded** batch-processing pipeline with secure file ingestion for PDF, DOCX, and ZIP uploads, processing 1K+ files per batch with fault-tolerant execution and cutting turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed API endpoints with pagination, field filtering, and caching for clients on **low-bandwidth networks**, improving response times 30% and cutting payload sizes across large-scale integrations used by distributed clients.
- Wrote 20+ automated unit and **integration test** suites with PyTest and Mockito, run nightly through Jenkins and GitHub-triggered branches across Windows, **Linux**, and macOS in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **C++** synchronization service on **Linux** that streamed real-time updates over TCP sockets, syncing 2M+ records across multiple sites with 33% fewer sync conflicts in a microservices platform.
- Automated **build and release** with Jenkins CI/CD, Docker, and staged rollouts, and debugged production issues from logs and metrics, cutting deployment time 50% with 95%+ successful build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built customer-facing Flask APIs for bookings, pricing, and coupons with configurable eligibility rules, keeping responses under 200ms across 50K+ records with Firebase real-time updates on Android, iOS, and web.
- Integrated multi-currency payment providers through idempotent webhooks for refunds and disputes, cutting manual handling about 70%, and implemented wallet-signature login that removed password storage for wallet users.

Skills

Languages: C++, C, Python, Java, JavaScript, TypeScript, SQL, Bash
Systems: Linux, Operating Systems, TCP/IP, Networking, Sockets, Multithreading, Concurrency
CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming, System Design, Debugging, Problem Solving
Build & Test: Git, CI/CD, Jenkins, Docker, Unit Testing, Integration Testing, PyTest, Mockito, Sentry, Kibana
Tools & Practices: AWS, Kubernetes, PostgreSQL, MySQL, GitHub Actions, Code Reviews, Documentation, Agile, Teamwork

Projects

Routing Table with Longest-Prefix Match | *C++, Linux, Data Structures, GoogleTest*

- Designed a **routing table** library that stores IPv4 and IPv6 prefixes in a compressed trie, performs **longest-prefix match** lookups, handles route add and withdraw, and benchmarks lookup speed.

BGP Route Selection Daemon | *C++, TCP Sockets, Linux, CLI*

- Designed a Linux daemon that keeps **BGP** sessions with peers over TCP, applies the best-path selection rules to learned routes, and exposes a **manageability** CLI for showing neighbors and routes.

CPU Scheduling Simulator Live | *C++, JavaScript, Operating Systems*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multi-level queue **scheduling algorithms**, testing preemptive and non-preemptive cases across 1K+ simulated process sets and reducing average turnaround 25%.